UNDERSTAND AI

Transformer

You've learned how AI turns text into tokens, and how each token's vector carries meaning. If only it were that easy.

Often a word's meaning isn't clear until you read the words around it. Here are three situations where a starting vector, on its own, falls short.

1 Different Meanings

`Please turn on the LIGHT so I can find my keys in the dark.`
`I packed only a few things so my bag would stay LIGHT for the hike.`

LIGHT has different meanings based on the words around it. If its starting vector locks in the not-heavy meaning, it's wrong for the first sentence, where LIGHT means brightness.

2 Pointing Words

`The trophy didn't fit into the suitcase because IT was too large.`
`The trophy didn't fit into the suitcase because IT was too small.`

Do you see how the sentence changes meaning, and we only changed the last word? In one case, **IT** means trophy. In the other, **IT** means suitcase.

3 Words Far Apart

`The cat sat on the mat during the May rainstorm because IT was tired.`

What does the word **IT** refer to: cat, mat, May, or rainstorm?

WHAT CAUSED THIS?

In each example above, the meaning of the word is only clear when you see the words around it. We know the meaning of **light** in the sentence "please turn on the **light**" because of the words "turn on." This was a foundational problem for AI, because it read text sequentially, one word at a time. Consider the sentence below. When AI read the word **IT**, the earlier words in the sentence began to fade.

How AI used to read

Information from early words had to travel down the chain, getting harder to hold onto with distance.

The → cat → sat → on → the → mat → during → the → May → rainstorm → because → it → was → tired

At first glance that seems fine. It's how humans read too. But we read with an advantage: we instantly know **IT** points back to **CAT**, even ten words later. A computer reading strictly in order doesn't, and that's what kept old AI from reading like we do.

THE BREAKTHROUGH

In 2017, eight researchers at Google published a paper called **Attention Is All You Need**. It introduced the **Transformer**, the architecture behind every modern LLM, and the "T" in ChatGPT. We'll share the specific concepts next, but for now, let's focus on the impact.

Instead of reading information sequentially, one word (token) at a time, the Transformer reads your whole text at once. That means it can establish the meaning of the word **IT** based on the words around it, like **CAT**. And, it sees that the **CAT** is **TIRED**.

It makes sense of all those words with two moves:

How AI reads now

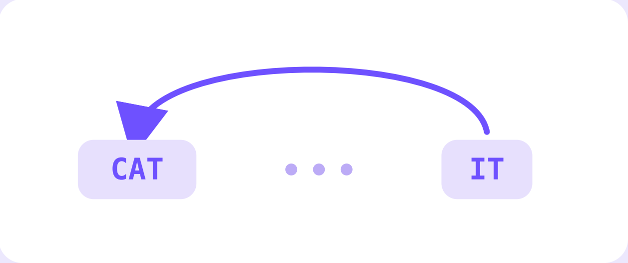
Any word can look directly at any other, no matter how far apart.

The cat sat on the mat during the May rainstorm because it was tired

AI does this in two steps:

1 Attention

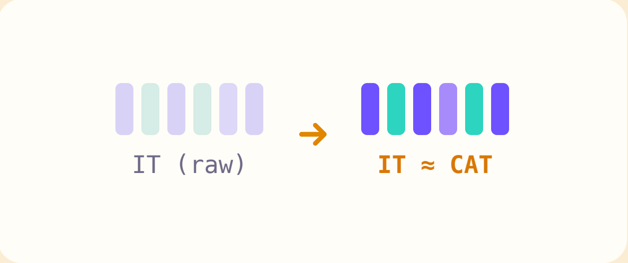
Which words matter?



Reading **IT**, the model weighs every word and leans hardest on **CAT**.

2 Transformation

Update the meaning



IT's vector updates to mean **CAT**.

Let's see how Attention and Transformation solved the old-AI problems we saw earlier in this lesson.

1 Different Meanings

`Please turn on the LIGHT so I can find my keys in the dark.`
`I packed only a few things so my bag would stay LIGHT for the hike.`

Attention: **LIGHT** links to "turn on" in the first sentence, to "bag" in the second.

Transformation: It sets **LIGHT**'s meaning: brightness in one, not-heavy in the other.

2 Pointing Words

`The trophy didn't fit into the suitcase because IT was too large.`
`The trophy didn't fit into the suitcase because IT was too small.`

Attention: "Too large" links **IT** to the trophy; "too small" links **IT** to the suitcase.

Transformation: It sets **IT**'s meaning: the trophy in one sentence, the suitcase in the other.

3 Words Far Apart

`The cat sat on the mat during the May rainstorm because IT was tired.`

Attention: **IT** leans most on **CAT** from ten words away; distance no longer weakens the link.

Transformation: It uses that link to update **IT**'s meaning, so the model knows **IT** is the **cat**.